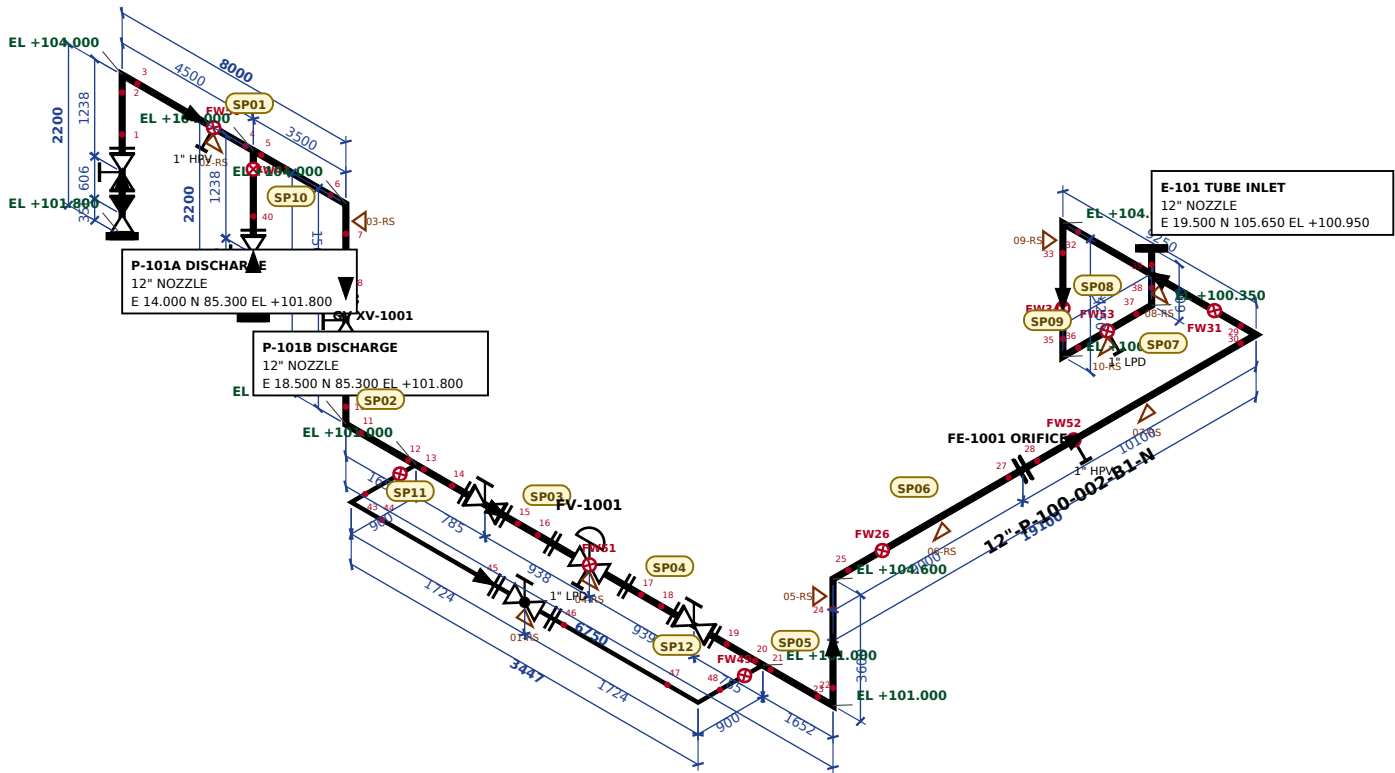
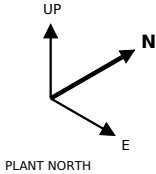


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore STD, ASTM A105	12"	11
2	PIPE STD, BE, ASTM A106 Gr.B SMLS	12"	43.8 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WPB	12"	9
4	TEE EQUAL, BW, STD, ASTM A234 WPB	12"	1
5	TEE RED., BW, STD, ASTM A234 WPB	12" x 8"	2
6	FLANGE WN RF CL300 B16.5, bore STD, ASTM A105 (orifice)	12"	2
7	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	8"	2.7 m
8	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	8"	2
9	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	8"	2
10	SOCKOLET 3000# (branch), ASTM A105	1"	4
11	ERECTION: CHECK VALVE SWING CL300 RF, A216 WCB, trim 8	12"	2
12	GATE VALVE OS&Y CL300 RF, A216 WCB, trim 8	12"	5
13	CONTROL VALVE FV-1001 CL300 RF (by I&C)	12"	1
14	ORIFICE PLATE + ORIFICE FLG PAIR FE-1001 (by I&C)	12"	1
15	GLOBE VALVE CL300 RF, A216 WCB, trim 8	8"	1
16	GASKET SPW 304/graphite, outer ring, CL300 B16.20	12"	17
17	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	12"	17
18	GASKET SPW 304/graphite, outer ring, CL300 B16.20	8"	2
19	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	8"	2
20	PIPE SUPPORTS (std. dwgs): rest x10	-	10

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	12"	401	SP01-2	12"	3789	SP01-3	12"	2789
SP01-4	12"	663	SP02-1	12"	663	SP02-2	12"	940
SP02-3	12"	151	SP03-1	12"	20	SP04-1	12"	20
SP05-1	12"	151	SP05-2	12"	940	SP05-3	12"	2686
SP05-4	12"	2150	SP06-1	12"	6264	SP07-1	12"	9514
SP07-2	12"	1350	SP08-1	12"	6986	SP08-2	12"	2150
SP09-1	12"	1186	SP09-2	12"	336	SP09-3	12"	14
SP10-1	12"	604	SP11-1	8"	341	SP11-2	8"	1029
SP12-1	8"	1029	SP12-2	8"	341			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=10.2m SP02=3.4m SP03=0.3m SP04=0.3m SP05=8.4m SP06=6.4m SP07=11.9m SP08=10.1m SP09=3.5m SP10=0.7m
SP11=2.1m SP12=2.1m

NOTES

- Pipe/fittings/flanges per class B1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 52.5 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Crude charge FV-1001 station at grade (EL 101.0) east of P-101B for operator access.
- Line passes between E-107 / E-108A and E-108B / E-108C at EL 104.6 (exchanger pipeway).
- E-101 bottom inlet only 0.95 m above grade - bottom connection run at EL 100.35 (see issues).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 12"-P-100-002-B1-N P-101A/B DISCHARGE TO E-101		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-006		REV A

LINE NUMBER 12"-P-100-002-B1-N	FROM P-101A/B	TO E-101	PIPING CLASS B1 - 300# CS ASTM A106 Gr.B
DESIGN P / T 35.0 barg / 60.0 C	OPER. P / T 23.99 barg / 30 C	TEST P (HYDRO) 52.5 barg (1.5xPxST/S=1.0)	PIPE WALL STD (9.53 mm)
INSULATION N 0 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 1 - rotating equipment (API 610 nozzle loads)
P&ID CFU-100-PR-PID-001	FLUID / PHASE P / L Crude charge pump discha	PAINT Epoxy/PU system	SPOOLS / WELDS 12 spools; 43 shop / 10 field